

Calvin Kwon

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EDUCATION

Cornell University

Bachelor's Degree in Mechanical Engineering

Graduation: May 2030

Ithaca, New York

Bergen County Technical High School - Teterboro

High School Diploma in Aerospace Engineering Academy

September 2022 - June 2026

Teterboro, New Jersey

PROJECTS AND AWARDS

Micro-gravity Nanolab

NASA HUNCH - National Design and Prototype Finalist

September 2025 - June 2026

- Lead team engineer: directed the design of a **2U Nanolab** payload to analyze zero-gravity fluid dynamics for agriculture in space
- Innovated an observation system and created **custom PCBs** throughout the design process to track water in microgravity and control electronics within the system; attended meetings with **NASA researchers** who gave regular insights to our prototypes
- Pitched the prototype and presented experimental findings at Rutgers University and developed/presented a refined prototype at a NASA HUNCH panel in The Johnson Space Center at Houston Texas

High-Altitude Microplastics Experiment

NASA Tchrise - National Student Challenge Winner

October 2024 - June 2025

- Selected as 1 of the 60 winning teams **out of 720+ nationally**, earning a **\$1,500 grant** to design, build, and launch a high-altitude microplastic collection experiment up to **70,000 ft** aboard a stratospheric balloon
- custom 3D-printed hardware, soldered sensor and motor circuits, and wrote CircuitPython code to control data collection and mechanical deployment, while receiving weekly technical mentorship from NASA engineers throughout the design and testing process
- Delivered a live virtual presentation summarizing flight data and experimental results to an audience of over **220 participants**

Underwater ROV

U.S. Navy SeaPerch - NY/NJ 3rd Place

January 2025 - April 2025

- Designed custom **3D-printed components**, **soldered motor control circuitry**, and collaborated with the team to prototype and iterate on frame designs optimized for speed during obstacle courses and stability while carrying payloads underwater
- Co-authored a comprehensive engineering **design portfolio** detailing ROV specifications, design rationale, and testing results, and piloted the vehicle through technical underwater obstacle and task courses during competition

EXPERIENCE

Mathematics Instructor

Mathnasium - The Math Learning Center

May 2026 - August 2026

Closter, New Jersey

- Math instruction from **Basic Mathematics through Pre-Algebra** to elementary and middle school students in small groups of 3-4
- Managed weekly 4-hour teaching shifts, adapting instructional strategies to support varying learning speeds and skill levels

Bergen Makerspace

Engineering and Design Intern

September 2025 - May 2026

Hackensack, New Jersey

- Engineered a fully functional, **1.5-meter wingspan RC plane** from scratch within a **\$300+ budget**, managing full-cycle CAD design, fabrication, and electronic integration
- Collaborated with a team to design and fabricate custom zoo enrichment tools, including a modular anteater tunnel system and an interactive feeder using **resin printing, CNC, and laser engraving**
- Educated local high school students on CAD tools and machinery operations while contributing 7 hours weekly to shop operations

Assistant Head Teacher

Kumon Math and Reading Center of Norwood

July 2025 - May 2026

Norwood, New Jersey

- Taught middle and high school students across **Pre-Algebra through Calculus BC**, providing personalized 1-on-1 and 1-on-2 instruction.
- Designed custom learning curricula and weekly homework schedules tailored to each student's pace and learning goals

Private Cello Tutor

Private Tutor

October 2025 - August 2026

Bergen County, New Jersey

- Delivered weekly **1-on-1** cello instruction to 5 high school students, preparing them for solo recitals, competitive orchestra auditions, and ensemble performances

SKILLS

Technical: Solidworks, Python, Autodesk Inventor (Certified), MATLAB, Onshape, Bambu, Laser Engraving/Cutting, MATLAB, Microsoft Office

Skills: Iterative Prototyping, Collaborative Teamwork, Technical Documentation, Systems Testing